

Project – Bookstagram

Progress Report-4

CSIS – 4495-002

Applied Research Project

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Work logs [After Progress report2]

Date	Hours	Description of Work Done
10/03/2026	4	Created a new Home page, merged profile page and dashboard page to one unified Dashboard page. Created layout for the story strip with new changes, and books section.
11/03/2026	4	Updated the story modal to working AI generated modal. Resolve error of generating images. Change the gemini's Api with hugging face's api to generate images. Resolved the save story error to view the story.
12/03/2026	2	Merged the 2 branch, added sidebar to profile page.
13/03/2026	3	Refactored stories functionality into new shared file. Applied some changes to the frontend of profile page pf the user
15/03/2026	2	Build follow and unfollow system.
16/03/2026	1	Writing a report

Summary Description of work done during this reporting period:

My primary focus during this reporting period was developing the AI-generated image story feature. Initially, I used the same Gemini API key to generate both the AI image stories and the eBook content. However, this approach resulted in multiple errors. To resolve this, I created a separate API key in Gemini Studio to handle the two functionalities independently. Despite this change, the image generation feature still did not work. After further investigation, I discovered that generating images with Gemini AI requires a paid API key. At that point, I began exploring alternative tools that could support free image generation. During my research, I found the Hugging Face platform, which provides APIs specifically for AI image generation. I integrated the Hugging Face API into the project and successfully generated images through AI. This was a significant milestone in implementing the AI-generated image story feature. In addition to this, I also began working on developing the follow and unfollow system.

Repo Check in of Implementation completed: The files and folders I have check in are as follows:

Branch: main

- Updated HomePage.jsx to include a styled "Go to Dashboard" button that navigates users to the DashboardPage using React Router's useNavigate hook.
- Merge branch 'main' of https://github.com/KhushiBhatt22/W26_4495_S2_KhushiB
- Merged profilepage and dashboardpage into one unified Dashboardpage. So the layout order is profile header, stories strip and all ebooks section
- Created a CreateStoryModal that opens directly on the dashboard when the user clicks the "Add Story +" button.
- Update Storymodal to generate story with AI, modified api end points from backend
- Resolved the image generation issue by switching from Gemini's paid image API to Hugging Face's free FLUX.1-schnell model, which successfully generates AI story images based on user prompts.
- Resolved the PayloadTooLarge Error by increasing Express's body size limit to 10mb in server.js, and fixed the Hugging Face token permissions to enable serverless Inference API calls, which allowed the Save Story functionality to work successfully.
- Resolved the error of showing general name when story is created by modifying storycontroller and DashboardPage files.
- Merge branch 'main' into Aditi_0044
- Updated DashboardPage to use the new sidebar layout instead of the old one. Updated some routes after the merge.
- Refactored the stories functionality into a shared StoriesStrip.jsx component, integrated it into both DashboardPage and NewDashboardPage, hid the topbar on DashboardPage

using a hideTopbar prop, and fixed misplaced hooks that were outside the component in NewDashboardPage.

- Build follow and unfollow system, update social controller, follow, unfollow endpoints

AI Usage –

AI Tool Name	Version / Account Type	Specific Feature for which AI Tool was Used	Value Addition (What value did you add beyond the AI?)
Claude	Sonnet 4.6/Free	To generate follow, unfollow endpoints	Created follow and unfollow endpoints to match the existing frontend state structure
Claude	Sonnet 4.6/Free	Debugged 500 error	Used browser console tab and network tab added console.error logging to isolate crash
Claude	Sonnet 4.6/Free	Suggested user wiring from mock to real backend	Identified dynamic gradient styling issue when replacing mock data
Claude	Sonnet 4.6/Free	To update DashboardPage to use NewDashboardLayout with sidebar	Identified the exact import swap and prop wiring (onCreateBook) needed to connect the sidebar's Create button to the existing modal state
Chatgpt	GPT 5.3/Free	Debugging gemini api image generation errors	Identified incorrect environment variable usage and fixed API key configuration in .env and controller
Chatgpt	GPT 5.3/Free	Debugging React frontend error when	Fixed issue where empty string was passed to img src

		rendering generated image	and implemented conditional rendering for generated images
Chatgpt	GPT 5.3/Free	Fixing backend logic for extracting generated images from Gemini API response	Adjusted code to correctly parse image data from the Gemini response and return Base64 image URL to frontend